

Sleep Apnea — Treatment & Complications



1. CPAP machine — first line

WHAT YOU SEE

A CPAP MACHINE sits on a shelf beside a GOLD STAR marked 'FIRST LINE' — the gold star flags CPAP as the first-line therapy for moderate-severe OSA.

WHAT IT ENCODES

Continuous positive airway pressure (CPAP) is the FIRST-LINE therapy for moderate-to-severe OSA (AHI ≥ 15 , or ≥ 5 with significant symptoms/comorbidity). It works as a pneumatic splint: continuous pressure delivered through a mask holds the collapsible pharynx open across the whole respiratory cycle, abolishing apneas/hypopneas, restoring oxygenation, and consolidating sleep. Auto-titrating (APAP) and fixed-pressure devices are options.

BOARD TRAP

WRONG answer: choosing an oral appliance or surgery first for moderate-severe OSA. CPAP is the most effective therapy and the first choice for these patients; alternatives are for mild disease or CPAP intolerance. The discriminator: CPAP is first-line; dental devices/surgery are second-line.

2. CPAP mask held up

WHAT YOU SEE

Man holding a CPAP mask in his hand

WHAT IT ENCODES

CPAP reliably improves daytime sleepiness (Epworth), neurocognitive function, quality of life, and lowers blood pressure (modestly, especially in resistant HTN). Its principal limitation is ADHERENCE — many patients struggle with the mask/pressure. Adherence is improved by good mask fit, humidification, treating nasal congestion, ramp features, and behavioral support; non-adherence is the most common reason CPAP 'fails.'

BOARD TRAP

WRONG answer: abandoning CPAP and jumping to surgery the moment a patient reports discomfort. The correct next step is to TROUBLESHOOT adherence (mask refit, humidification, pressure adjustment, desensitization) before switching to a less effective modality. The discriminator: most CPAP failure is an adherence problem, not a device-efficacy problem.

3. Weight loss — mild OSA

WHAT YOU SEE

'WEIGHT LOSS = MILD OSA' dumbbell tag

WHAT IT ENCODES

Weight loss is core therapy for all obese OSA patients and can be sufficient or even CURATIVE in mild, obesity-related disease. Losing parapharyngeal and visceral fat widens the airway and improves lung volumes; even 10% weight loss can substantially reduce AHI, and bariatric surgery markedly improves OSA in the morbidly obese. It complements CPAP in moderate-severe disease.

BOARD TRAP

WRONG answer: relying on weight loss ALONE in a patient with severe OSA who needs immediate symptom and risk control. Weight loss is slow and adjunctive in severe disease — CPAP is still needed now. The discriminator: weight loss can be definitive for MILD obesity-related OSA but is adjunctive for moderate-severe disease.

4. Oral appliance

WHAT YOU SEE

'ORAL APPLIANCE — MILD/MOD OSA' tag

WHAT IT ENCODES

Mandibular advancement devices protrude the lower jaw (and tongue) forward, enlarging the retroglossal airway. They are a reasonable option for MILD-to-MODERATE OSA, for CPAP-intolerant patients, or for predominantly positional/supine disease. They are less effective than CPAP, especially as severity rises, and require dental follow-up (bite changes, TMJ discomfort).

BOARD TRAP

WRONG answer: offering an oral appliance as primary therapy for severe OSA. Efficacy falls with severity, so CPAP remains first-line for AHI ≥ 30 . The discriminator: oral appliances are for mild-moderate disease or CPAP intolerance, not as first-line for severe OSA.

5. UPPP — less effective

WHAT YOU SEE

'UPPP = LESS EFFECTIVE, MILD/NON-OBESE' tag

WHAT IT ENCODES

Uvulopalatopharyngoplasty (UPPP) removes redundant soft palate/uvular/pharyngeal tissue but has variable, often modest and non-durable success because OSA collapse is usually multilevel. It is reserved for selected patients (clear anatomic obstruction, non-obese, CPAP failure). Other surgical options: tonsillectomy (esp. children), maxillomandibular advancement (most effective surgery in adults), and hypoglossal nerve stimulation in selected CPAP-intolerant patients.

BOARD TRAP

WRONG answer: recommending UPPP as the first-line treatment for typical adult OSA. Surgery is NOT first-line — CPAP is. UPPP in particular is unreliable in obese, multilevel disease. The discriminator: surgery is a salvage/selected-anatomy option after CPAP failure, not initial therapy.

6. Avoid sedatives/alcohol

WHAT YOU SEE

Crossed-out bottle 'NO FDA DRUG' / avoid agents area

WHAT IT ENCODES

Alcohol, benzodiazepines, opioids, and other sedative-hypnotics must be AVOIDED: they depress the pharyngeal dilator muscles and the arousal/ventilatory response, prolonging apneas, deepening desaturations, and worsening OSA. Alcohol near bedtime acutely increases AHI. Opioids can additionally precipitate or worsen CENTRAL sleep apnea.

BOARD TRAP

WRONG answer: prescribing a benzodiazepine or 'sleeping pill' to a snoring, sleepy patient for 'insomnia' before evaluating for OSA. Sedatives worsen apneas and can be dangerous in undiagnosed OSA. The discriminator: in suspected/known OSA, sedatives and alcohol are contraindicated/avoided — they aggravate airway collapse.

7. No FDA drug

WHAT YOU SEE

A crossed-out pill barrel labelled 'NO FDA DRUG' — the red X over the drug barrel means no medication treats the airway obstruction itself.

WHAT IT ENCODES

No medication reliably treats the airway obstruction of OSA itself — management is mechanical (CPAP/BiPAP, oral appliance) and behavioral (weight loss, positional therapy, avoiding sedatives). Wake-promoting agents (modafinil/armodafinil, solriamfetol) can be ADJUNCTS for residual daytime sleepiness despite adequate CPAP use, but they treat the symptom, not the apneas.

BOARD TRAP

WRONG answer: selecting a pharmacologic agent (e.g., modafinil) as PRIMARY therapy for OSA. Wake-promoting drugs only address leftover sleepiness in already-treated, adherent patients — they do nothing for the obstruction. The discriminator: primary OSA therapy is mechanical/behavioral; drugs are at most adjuncts for residual sleepiness.

8. Resistant HTN cuff

WHAT YOU SEE

BP cuff with spikes, 'SYMPATHETIC NERVOUS SYSTEM → BLOOD PRESSURE SPIKES EACH ARM CUFF'

WHAT IT ENCODES

Each apnea triggers hypoxia and arousal that drive sympathetic surges, raising blood pressure repeatedly through the night and chronically. OSA is a leading cause of SECONDARY and RESISTANT hypertension and of the non-dipping nocturnal BP pattern. CPAP produces a modest but real BP reduction, greatest in resistant hypertension.

BOARD TRAP

WRONG answer: escalating to a fourth or fifth antihypertensive in an obese snorer with resistant HTN without screening for OSA. OSA is a treatable secondary cause that should be evaluated. The discriminator: resistant hypertension (≥ 3 drugs incl. diuretic) plus snoring/sleepiness/obesity warrants a sleep study, not just more pills.

9. Non-dipping → secondary HTN

WHAT YOU SEE

'NON-DIPPING = SEVERE OSA' tag

WHAT IT ENCODES

Normally blood pressure 'dips' $\geq 10\%$ during sleep; OSA blunts this, producing a NON-DIPPING (or reverse-dipping) pattern on ambulatory BP monitoring, a marker of more severe disease and elevated cardiovascular/target-organ risk. Non-dipping plus daytime hypertension in an obese snorer is a strong pointer to OSA.

BOARD TRAP

WRONG answer: ignoring a non-dipping pattern on ABPM as 'monitoring artifact.' Loss of nocturnal dipping is a meaningful sign that should prompt OSA evaluation. The discriminator: non-dipping/resistant nocturnal hypertension is a recognized OSA fingerprint and a cardiovascular risk marker.

10. AF / arrhythmia trophy

WHAT YOU SEE

Fallen trophy with 'A-FIB' / arrhythmia tag

WHAT IT ENCODES

OSA strongly predisposes to atrial fibrillation (via atrial stretch from negative intrathoracic pressure swings, hypoxia, sympathetic surges, and inflammation) and to nocturnal bradyarrhythmias/AV block during apneas. Untreated OSA markedly increases AF RECURRENCE after cardioversion or ablation; treating OSA with CPAP improves rhythm-control success.

BOARD TRAP

WRONG answer: attempting repeat cardioversion/ablation in recurrent AF without screening for and treating coexisting OSA. Untreated OSA undermines rhythm control. The discriminator: recurrent AF after rhythm-control intervention should prompt evaluation for OSA, whose treatment improves outcomes.

11. Pulmonary HTN → right heart failure

WHAT YOU SEE

Crossed-out heart 'RIGHT HEART FAILURE' figure at right

WHAT IT ENCODES

Chronic nocturnal hypoxemia causes hypoxic pulmonary vasoconstriction and vascular remodeling, producing pulmonary hypertension and, over time, cor pulmonale (right heart failure) with peripheral edema, elevated JVP, and a loud P2. This is WHO Group 3 pulmonary hypertension (lung/hypoxia-related). Treating the OSA (and any coexisting obesity hypoventilation) is central to management.

BOARD TRAP

WRONG answer: working up isolated right-sided heart failure/pulmonary hypertension in an obese, sleepy snorer without a sleep study. OSA (and obesity hypoventilation) is a reversible contributor to Group 3 pulmonary hypertension. The discriminator: nocturnal-hypoxia-driven pulmonary HTN points to a sleep-breathing disorder as the cause.

12. Obesity hypoventilation (BiPAP)

WHAT YOU SEE

Obese man at door with 'IPAP/EPAP — BIPAP = 2 PRESSURES, CO2 RETENTION' panel

WHAT IT ENCODES

Obesity hypoventilation syndrome (OHS, 'Pickwickian') = obesity (BMI ≥ 30) + AWAKE DAYTIME hypercapnia (PaCO₂ ≥ 45 mmHg) not explained by another cause; most OHS patients also have OSA. Treat with BiPAP, which delivers two pressures — higher inspiratory (IPAP) and lower expiratory (EPAP) — so the IPAP–EPAP difference provides ventilatory support to blow off CO₂, unlike plain CPAP (one pressure) which only splints the airway.

BOARD TRAP

WRONG answer: prescribing plain CPAP for a hypercapnic obese patient with daytime CO₂ retention. CPAP alone may not correct the hypoventilation — BiPAP (or volume-targeted support) is preferred when daytime hypercapnia is present. The discriminator: OHS is defined by AWAKE daytime hypercapnia (vs simple OSA, which has normal daytime CO₂), and it needs bilevel ventilatory support.

13. Mortality benefit unproven

WHAT YOU SEE

A trophy with a question-mark tag 'NO CV MORTALITY BENEFIT?' under investigation — the questioned trophy means CPAP's hard cardiovascular-mortality benefit is unproven.

WHAT IT ENCODES

CPAP clearly improves daytime sleepiness, quality of life, and blood pressure, and reduces accidents. However, large randomized trials (e.g., SAVE) did NOT demonstrate a clear reduction in hard cardiovascular events or mortality — partly attributed to low CPAP adherence in trials. CPAP is still indicated for symptom control and risk-factor modification.

BOARD TRAP

WRONG answer: claiming CPAP has 'proven mortality benefit' as the main reason to prescribe it, or conversely refusing CPAP because trials were neutral on mortality. The evidence-based justification is symptomatic and BP/quality-of-life benefit (and accident reduction). The discriminator: CPAP's documented benefits are symptomatic/QoL/BP, while a hard cardiovascular mortality benefit remains unproven.
